

AI Architecture Decision Guide

Score any AI use case across 5 dimensions to find the right architecture.

Elliot Betancourt

Teams pick AI architecture by mimicry or enthusiasm. Then six months in, they're maintaining something more expensive, more fragile, and harder to explain than what they replaced. This framework changes the decision from intuition to criteria.

1 Formality of Output — Can you verify correctness programmatically?

5 = code/math (run it and know) 3 = structured output with clear criteria 1 = pure judgment call

Formal outputs — code, SQL, structured data — can be tested. Judgment outputs cannot. Agents need verifiability to close the feedback loop.

2 Feedback Loop Speed — How fast do you know if it worked?

5 = seconds (automated test or check) 3 = hours/days (manual review) 1 = weeks/months (downstream impact)

Fast feedback enables iteration and self-correction. Slow feedback turns every agent failure into a deferred liability.

3 Task Repetition — How often is this exact task run?

5 = hundreds per day (high volume) 3 = weekly/monthly cadence 1 = one-off or ad-hoc

Automation ROI compounds with repetition. One-off tasks rarely justify the overhead of building and maintaining an agent.

4 Context Complexity — Is context well-defined?

5 = fully specified inputs and outputs 3 = mostly explicit, some interpretation 1 = requires implicit institutional knowledge

Agents work from what you give them. If the context lives in someone's head, you're not ready to automate — you're ready to document.

5 Error Cost — What happens when it fails?

5 = easy to catch and revert 3 = visible error, recoverable 1 = irreversible or customer-facing damage

High error cost demands human review. Low error cost enables full automation. Know which world you're in before you ship.

Decision Matrix

Total Score	Recommendation
21-25	Full agent automation — all five signals are favorable.
15-20	Agent-assisted — human reviews final output.
10-14	Tools not agents — deterministic workflows, not generative.
Below 10	Keep human — judgment, stakes, or context complexity makes automation a liability.

The bottleneck is not the model. It is knowing where to point it.